Enterprise Blockchain Case Studies

TypeScript repository demonstrating enterprise blockchain patterns.

Focus areas

- Operational case studies (traceability, privacy, credentials, settlement)
- Protocol adapters (Fabric, Besu, Corda)
- Off-chain cryptography (MPC, HSM, post-quantum)

Repository Structure

Folder	Contents
`modules/`	Domain logic, protocol adapters, integration clients
`examples/`	20 runnable scenarios
`contracts/`	Solidity (Foundry), Fabric chaincode (TS), Corda (Kotlin)
`docs/`	Architecture decisions, flow diagrams, this deck
`skills/`	AI skill files for assisted development

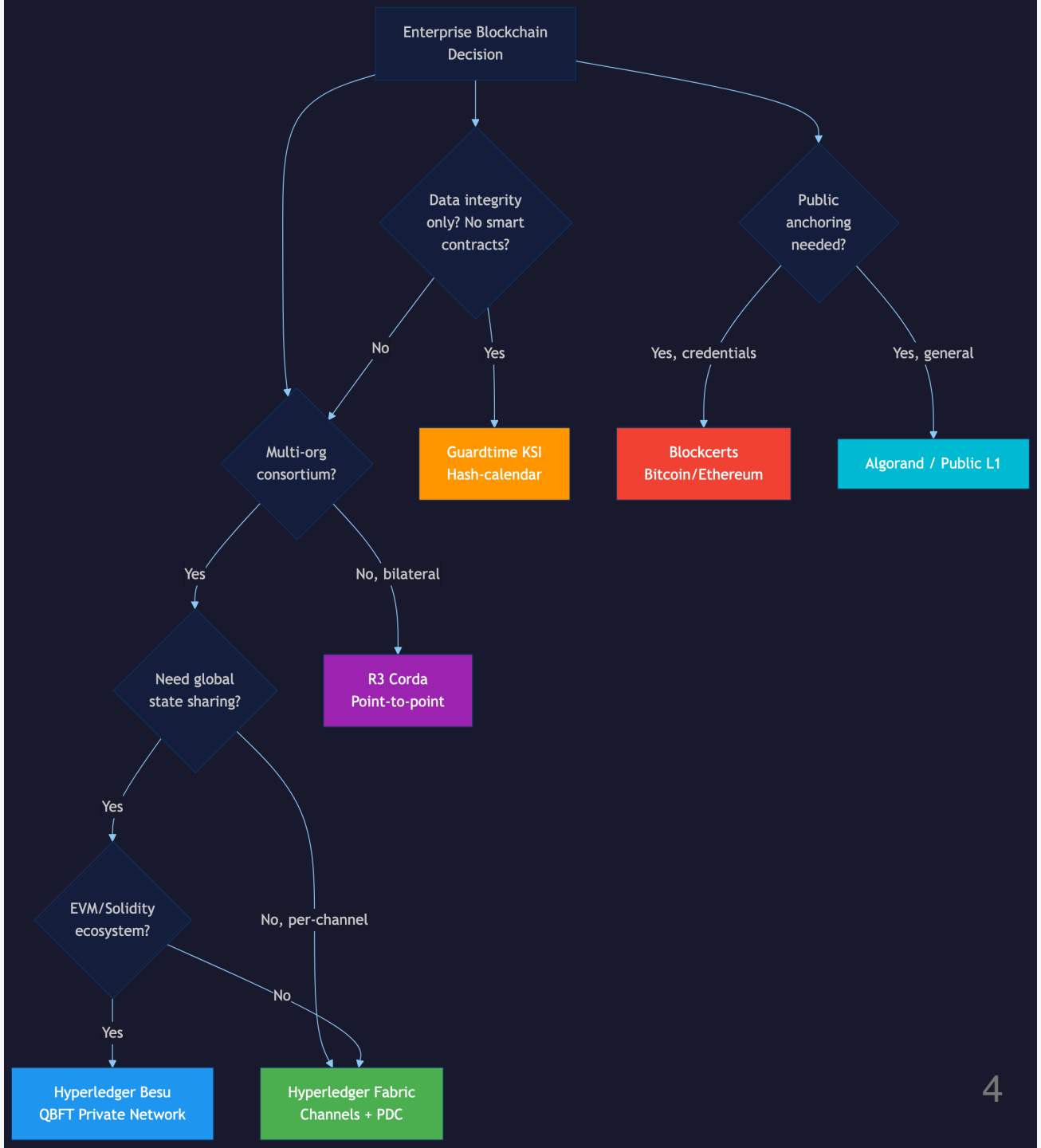
Case Studies

Scenario	Domain Problem	Protocol Fit
Food recall	Trace contaminated lots across supply chain	Fabric (channel isolation)
Order sharing	Selective disclosure to bank/logistics/regulator	Besu (privacy groups)
Staffing clearance	Verify credentials before clinical assignment	Corda (point-to-point)
Aid reconciliation	Multi-agency voucher settlement	Besu (shared state)

Platform Selection

Decision tree for protocol choice:

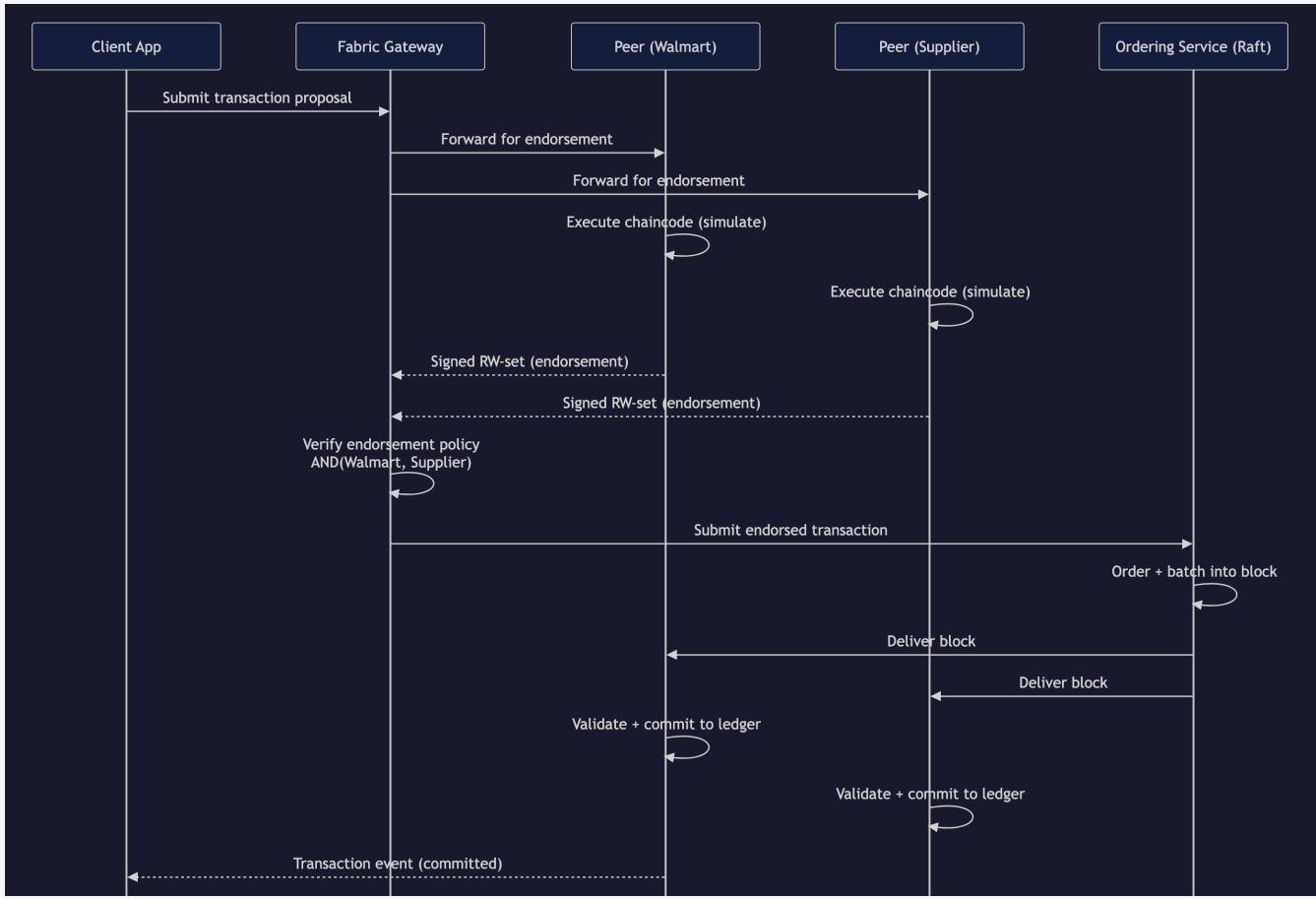
- Data integrity only → **KSI / Guardtime**
- Public anchoring → **Bitcoin / Ethereum**
- Bilateral flows → **Corda**
- EVM ecosystem → **Besu**
- Channel isolation → **Fabric**



Fabric: Endorsement Flow

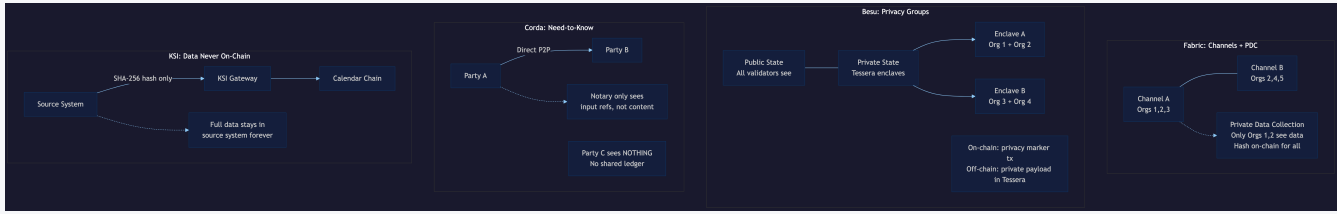
1. Client submits proposal
2. Endorsing peers simulate chaincode
3. Client collects endorsements
4. Orderer batches into block
5. All peers validate and commit

Use case: **Food recall traceability**



Privacy Patterns

Platform	Mechanism
Fabric	Private data collections, transient data
Besu	Privacy groups, restricted contract state
Corda	Need-to-know state distribution

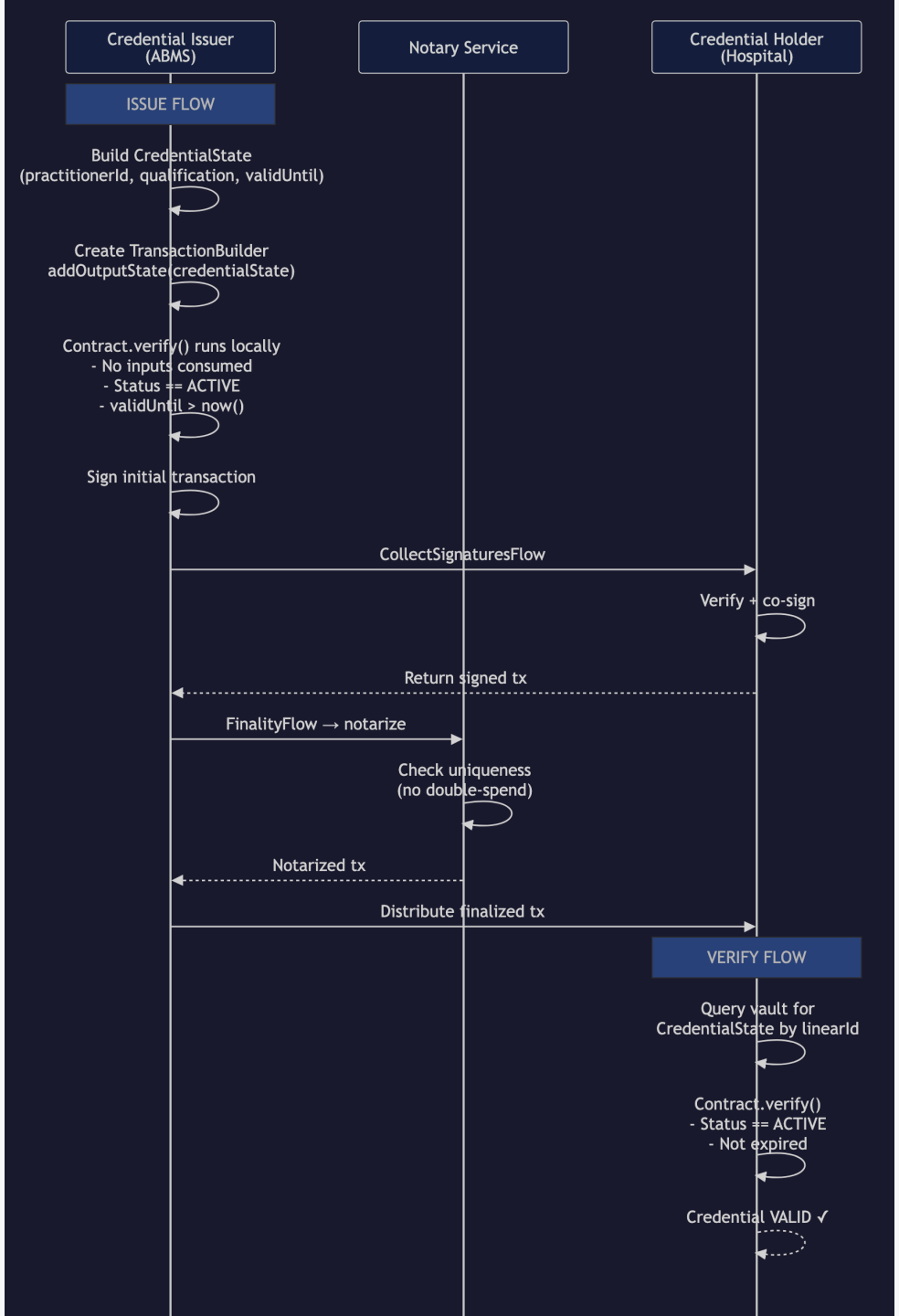


Use case: **Consortium order sharing**

Corda: Flow Protocol

1. Issuer builds `CredentialState`
2. Contract verifies locally
3. `CollectSignaturesFlow` gathers counterparty signatures
4. Notary checks uniqueness
5. `FinalityFlow` distributes to participants

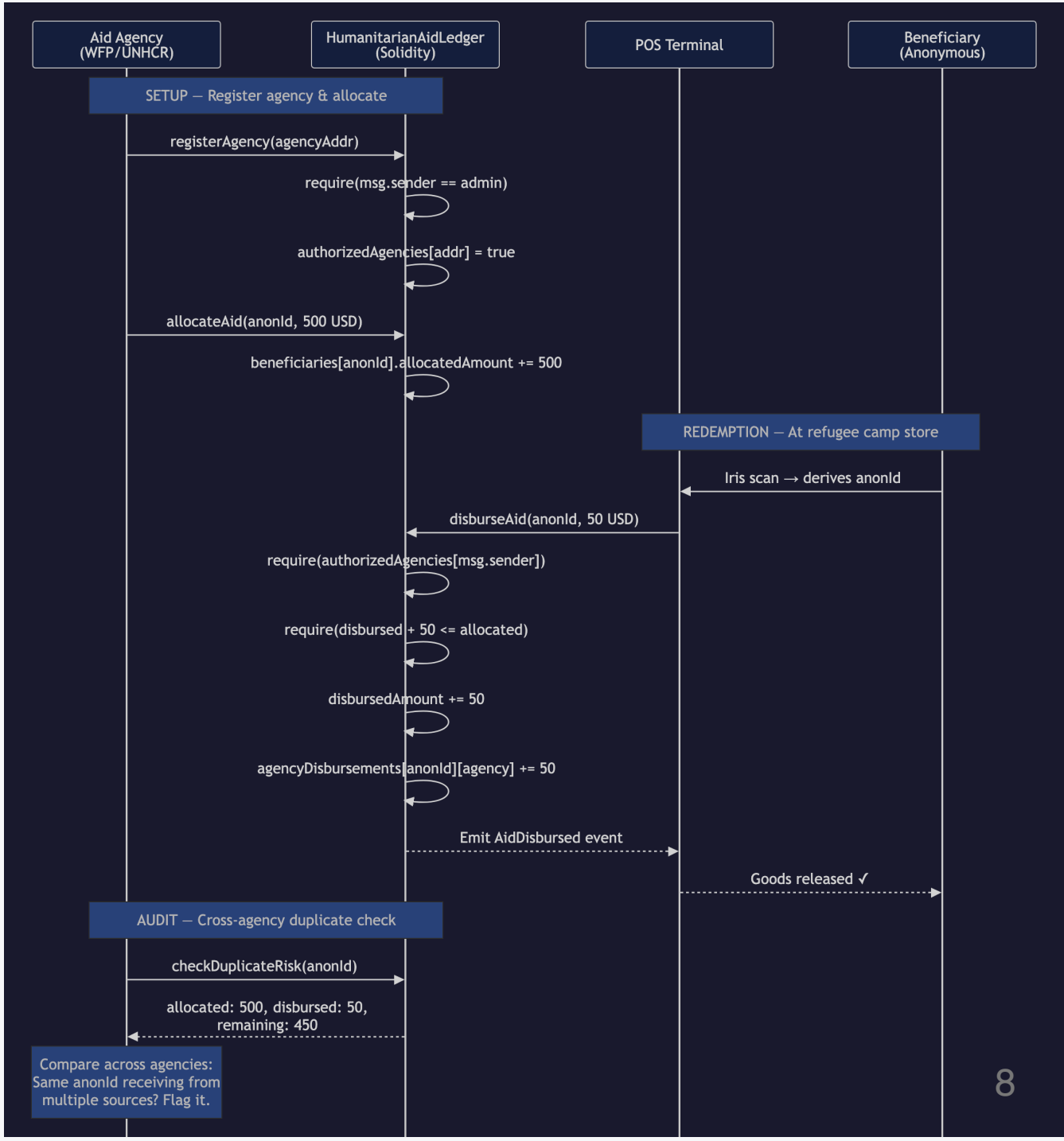
Use case: **Hospital staffing clearance**



Settlement Controls

- Agency registers and allocates funds
- Beneficiary redeems at POS
- Contract enforces budget limits
- Cross-agency duplicate detection

Use case: **Aid voucher reconciliation**



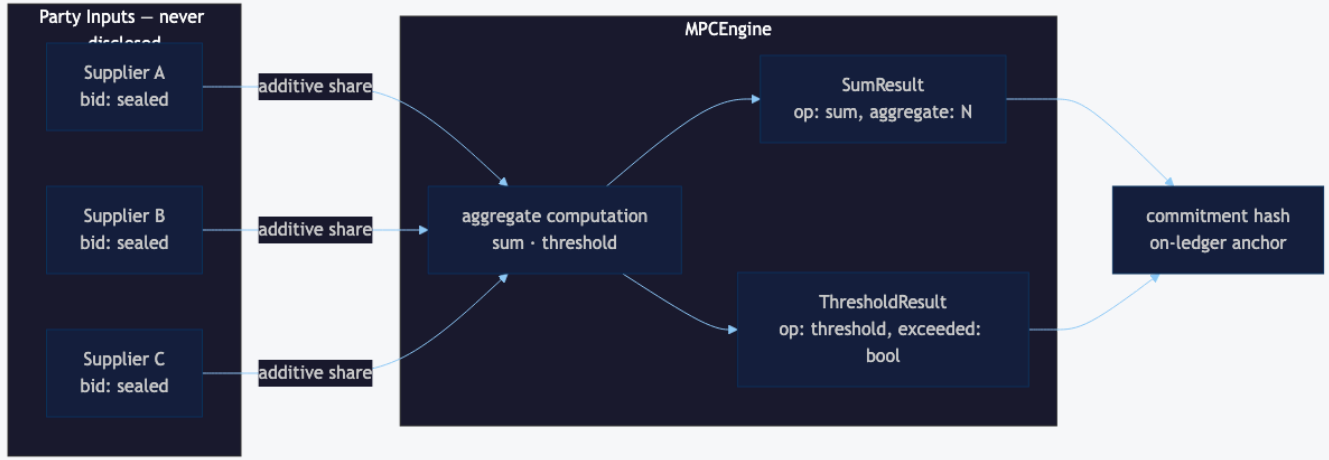
MPC: Additive Secret Sharing

Parties split inputs into additive shares:

$$\text{secret} = \text{share}_1 + \text{share}_2 + \text{share}_3 \pmod p$$

Computation happens on shares; result reconstructed.

Examples: `mpc-sealed-bid-auction`, `mpc-joint-risk-analysis`



MPC: Threshold Schemes

Shamir Secret Sharing (k-of-n)

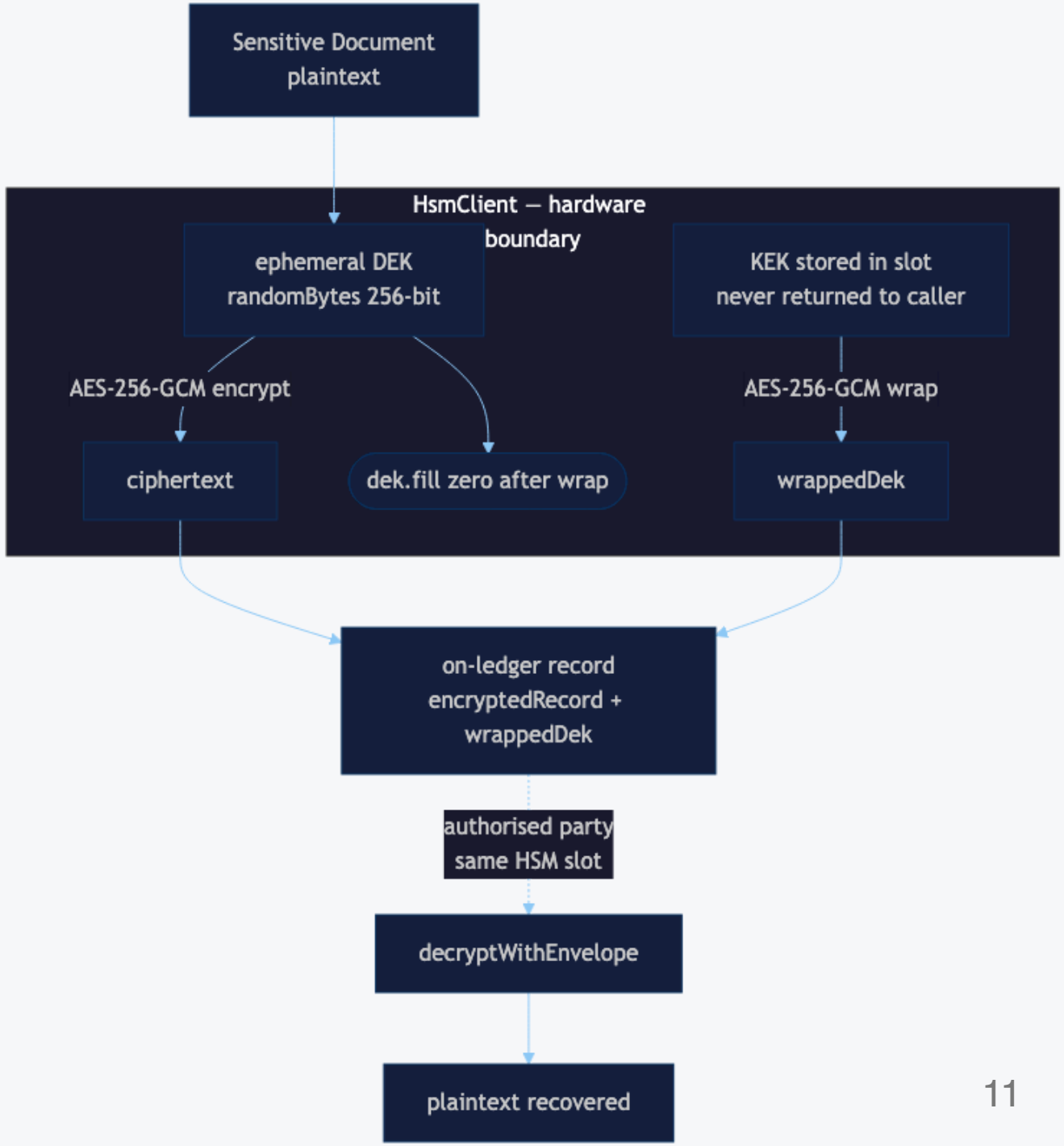
- Polynomial interpolation over finite field
- Any k shares reconstruct; k-1 reveal nothing
- Used for key custody and quorum authorization

Example: `quantum-resistant-key-sharing` — 3-of-5 threshold + hash-ladder anchoring

HSM: Envelope Encryption

1. Generate ephemeral DEK (256-bit)
2. Encrypt payload with DEK (AES-GCM)
3. Wrap DEK with KEK inside HSM
4. Store `ciphertext + wrappedDEK`
5. Only HSM can unwrap

Private keys never leave hardware boundary.



HSM Examples

Example	Pattern
`hsm-transaction-signing`	EC P-256 ECDSA with audit digest
`hsm-key-ceremony`	3-of-5 Shamir + HSM-signed root certificate
`hsm-envelope-encryption`	DEK/KEK for on-ledger document confidentiality

Post-Quantum Cryptography

NIST Standards (August 2024)

Standard	Algorithm	Wire Sizes
FIPS 203	ML-KEM-768	pk: 1184 B, ct: 1088 B, ss: 32 B
FIPS 204	ML-DSA-65	pk: 1952 B, sig: 3309 B

Replaces RSA/ECDH (key exchange) and ECDSA (signatures).

ML-KEM Key Encapsulation

Lattice-based KEM (FIPS 203)

1. Recipient generates `(pk, sk)`
2. Sender: `(ciphertext, sharedSecret) = encapsulate(pk)`
3. Recipient: `sharedSecret = decapsulate(sk, ciphertext)`

Parameter sets: ML-KEM-512, ML-KEM-768, ML-KEM-1024

Example: `kyber-kem-key-exchange`

Hybrid KEM

X25519 + ML-KEM-768

```
combinedSecret = HKDF(x25519Secret || mlkemSecret)
```

- Defense-in-depth: secure if either primitive holds
- Recommended for transition period
- Backward compatible with classical peers

Example: `hybrid-kem-settlement`

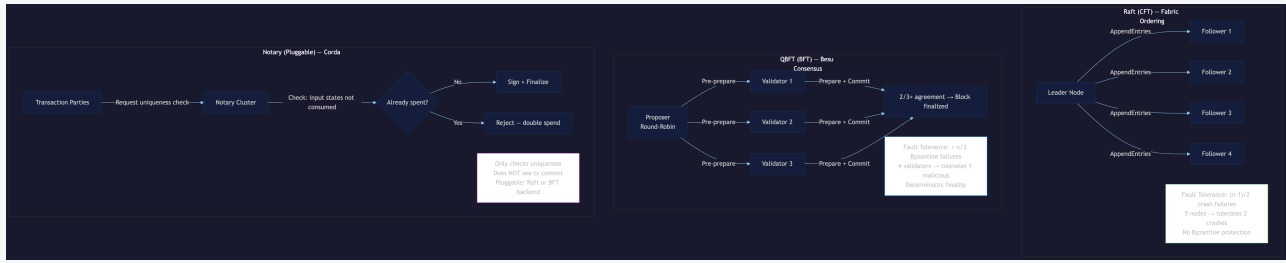
Quantum-Safe Payment Flow

Phase	Primitive
Key ceremony	Hybrid KEM (X25519 + ML-KEM-768)
Sign instruction	ML-DSA-65 (FIPS 204)
Encrypt payload	AES-256-GCM with hybrid key
Authorize settlement	3-of-3 MPC threshold

Example: `quantum-safe-payment`

Consensus Comparison

Protocol	Consensus	Finality
Fabric	Raft / BFT	Immediate
Besu	QBFT / IBFT 2.0	Immediate
Corda	Notary (single/clustered)	Immediate



All provide deterministic finality (no forks).

Architecture Layers

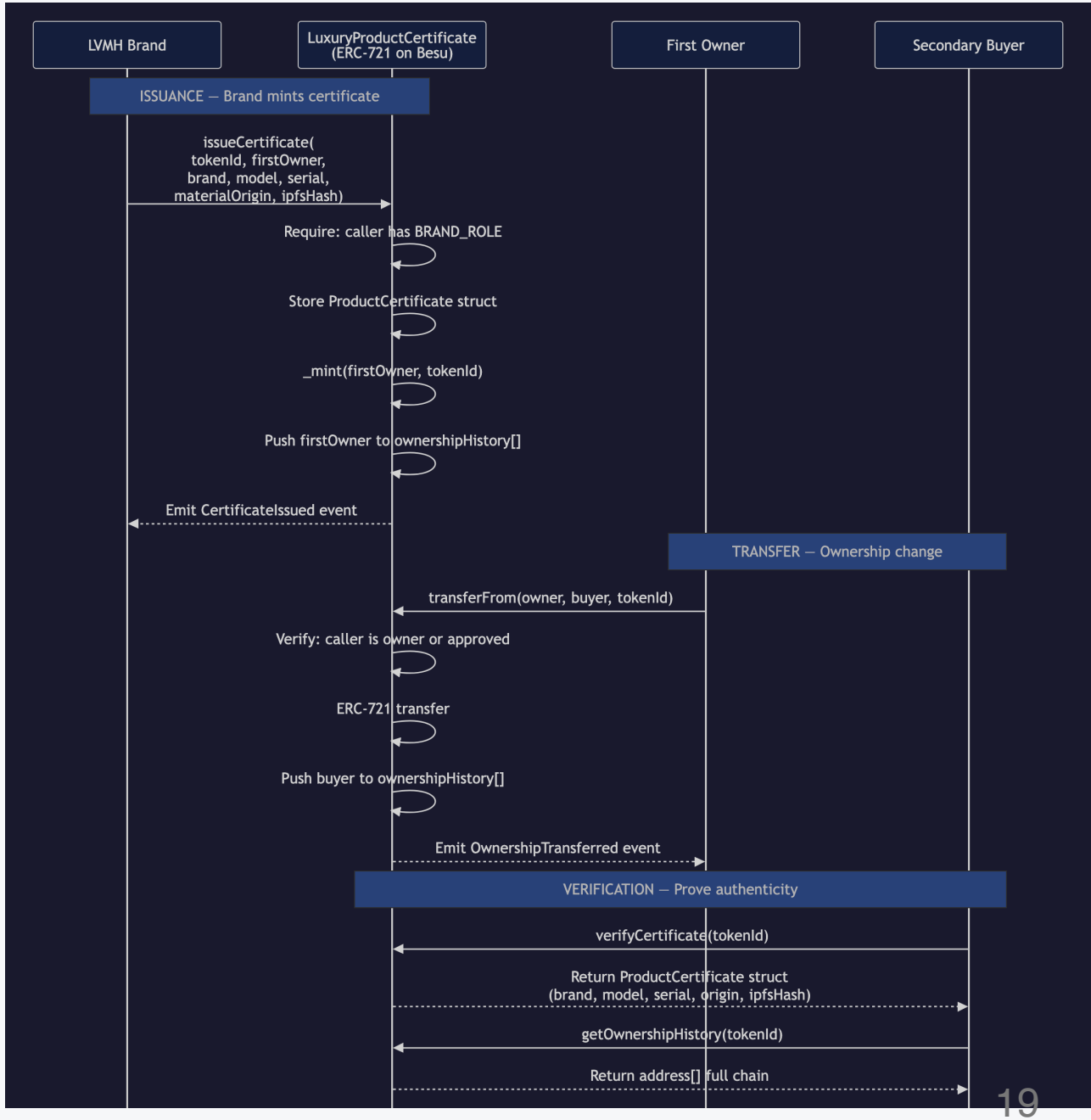
Layer	Responsibility	Location
Domain	Business rules, entities	`modules/{domain}/`
Protocol	Platform-specific tx shapes	`modules/protocols/`
Integration	SDK bindings, retry, errors	`modules/integrations/`
Config	Endpoints, credentials	`examples/config/`

Besu Integration

Stack: Solidity + ethers.js + privacy groups

Repository assets:

- `contracts/solidity/src/` — Foundry project
- `modules/integrations/besu-client/` — tx builders
- `modules/protocols/besu/` — privacy group calls

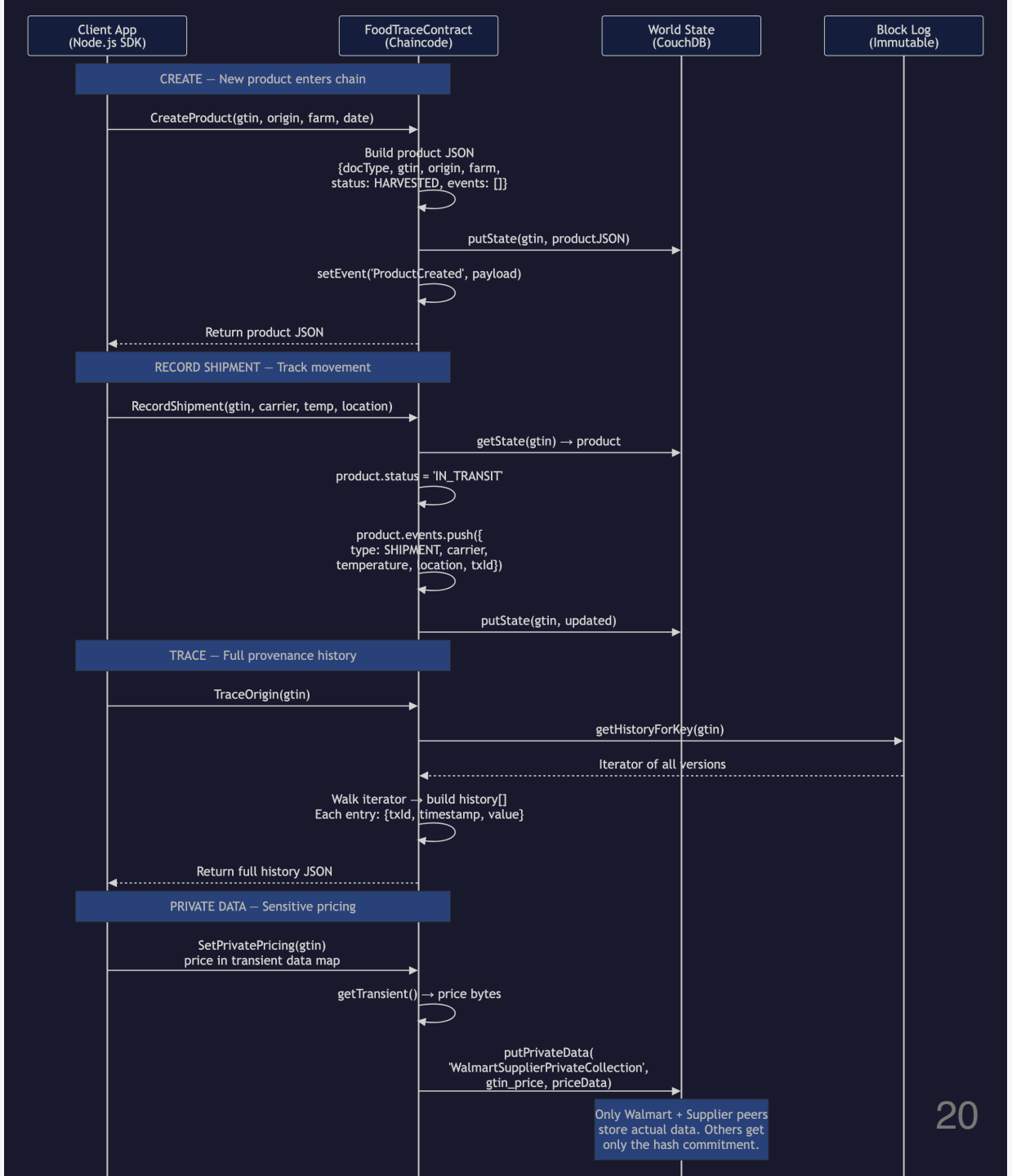


Fabric Integration

Stack: TypeScript chaincode + Gateway SDK

Repository assets:

- `contracts/fabric/` — FoodTraceContract
- `modules/integrations/fabric-gateway/` — proposal builders
- `modules/protocols/fabric/` — invoke/query shapes



Corda Integration

Stack: Kotlin CorDapp + REST gateway

Repository assets:

- `contracts/corda/` — ProviderClearanceContract/Flow/State
- `modules/integrations/corda-gateway/` — HTTP request builders
- `modules/protocols/corda/` — flow invocation shapes

TypeScript handles the integration boundary, not CorDapp runtime.

Quick Start

```
npm install           # install dependencies
npm run verify        # format + lint + typecheck + test + examples
npm run demo:adapters       # protocol adapter projections
npm run demo:integrations   # SDK integration sketches
```

All examples run offline with mocked SDK boundaries.

Example Commands

Category	Commands
Case studies	`example:food-recall`, `example:order-sharing`, `example:staffing-clearance`, `example:aid-reconciliation`
Protocol	`example:fabric-projection`, `example:besu-projection`, `example:corda-projection`
Integration	`example:fabric-gateway`, `example:besu-ethers`, `example:corda-rest`
MPC	`example:mpc-auction`, `example:mpc-risk-analysis`, `example:quantum-key-sharing`
HSM	`example:hsm-tx-signing`, `example:hsm-key-ceremony`, `example:hsm-envelope-encryption`
PQC	`example:kyber-kem`, `example:hybrid-kem`, `example:quantum-safe-payment`

Navigation

Goal	Start
Run a scenario	`examples/`
Read domain logic	`modules/{domain}/src/`
See protocol shapes	`modules/protocols/`
Study integration patterns	`modules/integrations/`
Review architecture	`docs/architecture/`

Summary

- **4** case studies (traceability, privacy, credentials, settlement)
- **3** protocol adapters (Fabric, Besu, Corda)
- **3** integration clients (Gateway, ethers, REST)
- **9** cryptographic examples (3 MPC + 3 HSM + 3 PQC)

All pass `npm run verify` and run offline.